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CENG 408 – Testing & Validation Report

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Authors

Orkun Oğuztürk-202111078

Can Berk Meşe-202111045

Ömer Faruk Şahin-202111073

Barbaros Murat Dönmez-202011019

Furkan Çoban-202011204

Arda Kaan Bakır-202111064

Elif Güngör-202111077

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1.Introduction

This testing and validation report presents the evaluation of a multi-stage lung nodule analysis pipeline developed on the LIDC-IDRI dataset. The proposed system integrates anatomically aware preprocessing, ROI-based 2.5D segmentation, knowledge distillation-based classification, and Grad-CAM-based interpretability into a single workflow for AI-assisted lung nodule analysis. The overall objective of this report is to examine whether these components operate correctly, produce reliable outputs, and provide interpretable results across data preparation, training, inference, and visual explanation stages.

The validation process focuses on several key questions. First, it evaluates whether CT and SEG data are matched correctly and whether reconstructed masks remain anatomically consistent with CT slices after preprocessing. Second, it examines whether the segmentation pipeline can generate visually coherent and learnable ROI-based 2.5D samples. Third, it assesses whether the knowledge distillation framework successfully transfers useful volumetric information from the 3D teacher model to the lighter 2.5D student model. Finally, it investigates whether Grad-CAM visualizations provide meaningful spatial evidence about the regions influencing model predictions.

Accordingly, the report evaluates the system in terms of technical correctness, predictive performance, interpretability, and current limitations. This evaluation is supported by both quantitative and qualitative evidence, including Dice score, accuracy, loss trends, training progression, representative prediction outputs, probability maps, and Grad-CAM activation visualizations. In this way, the report aims to provide a structured validation basis for the current pipeline and to identify the main areas that require further improvement.

2. System Overview

The evaluated system is designed as a multi-stage lung nodule analysis pipeline that integrates data preparation, segmentation, classification, knowledge distillation, and visual interpretation within a single workflow. The main purpose of this system is to transform raw medical imaging data into anatomically consistent model inputs, train models that can learn nodule related spatial patterns, and support the validation of model outputs through visual interpretation.

The first component of the system is the preprocessing module. In this stage, raw LIDC-IDRI CT and SEG data are prepared for model training and validation. CT and SEG series are matched, segmentation masks are reconstructed in CT slice space, nodule-centered ROI regions are extracted, and 2.5D samples are generated from consecutive CT slices. These preprocessing outputs provide a common input foundation for the later segmentation and classification stages.

The second component is the ROI-based 2.5D segmentation module. This module uses the preprocessed 2.5D image samples together with their corresponding reconstructed masks to learn pixel-level nodule localization. The purpose of this stage is to evaluate whether the model can identify nodule regions in anatomically aligned ROI-based inputs.

The third component is the knowledge distillation-based classification structure. In this framework, a 3D teacher model learns volumetric nodule-related features from 3D patches, while a lighter 2.5D student model is trained to approximate this knowledge using 2.5D inputs. This design allows the system to preserve part of the spatial information captured by the 3D model while reducing computational cost and supporting more efficient inference. Unlike the segmentation module, this component produces a classification output rather than a pixel-level mask.

The final component is the Grad-CAM-based interpretability module. This module is used to visualize the spatial regions that most strongly influence model predictions. By presenting the CT slice, the expected nodule region, and the activation map together, the module supports visual inspection of whether model attention is concentrated on clinically meaningful areas and helps make the decision process more transparent.

Overall, the workflow of the system can be summarized as follows: raw CT and SEG data are transformed into anatomically aligned ROI-based samples; segmentation and classification components are trained using these prepared inputs; and the resulting model outputs are examined through visual interpretation. In this way, the pipeline brings together preprocessing, prediction, and interpretability in an integrated validation-oriented framework..

3. Testing Scope and Validation Criteria

This section defines the scope of evaluation for the developed system and the main criteria used to validate each component. The purpose is not only to show that the system can be executed, but also to examine whether it operates in a technically consistent, reliable, and interpretable manner throughout preprocessing, training, inference, and visual explanation stages.

3.1 Preprocessing Validation Criteria

In the preprocessing stage, validation is based on whether raw CT and SEG data are transformed into anatomically correct and learnable samples suitable for later model training and inference. The main criteria include the correct matching of CT and SEG data, the spatial consistency of reconstructed masks with CT slices, the ability of ROI extraction to cover the lesion region while reducing irrelevant surrounding content, and the consistency of 2.5D sample generation between the center slice and its neighboring slices. In addition, preserving mask alignment with the target center slice after resizing and sampling is considered a key validation requirement. Since

preprocessing also forms the basis of large-scale experiments, the computational efficiency of sample generation is included as an additional validation criterion.

3.2 Segmentation Validation Criteria

In the evaluation of the segmentation module, the main objective is to determine whether the model can learn meaningful spatial representations from the generated ROI-based 2.5D samples and produce clinically reasonable mask predictions. The validation criteria for this component include training and validation Dice scores, loss trends across epochs, regional overlap between predicted masks and ground-truth masks, and visual consistency in representative validation outputs. In addition, the evaluation considers whether the model can capture the main lesion area while avoiding boundary leakage and small false-positive regions. The sustainability and reproducibility of the training process are also treated as indirect validation criteria through the checkpoint-based saving mechanism.

3.3 Classification / Distillation Validation Criteria

The main goal in validating the knowledge distillation-based classification structure is to assess whether the volumetric information learned by the 3D teacher model is effectively transferred to the lighter 2.5D student model. Therefore, the validation criteria include the teacher model demonstrating stable and reliable learning, the student model improving throughout training, classification accuracy increasing across epochs, and distillation loss showing a decreasing trend. It is also evaluated to what extent the student model can preserve useful spatial information while operating at a lower computational cost than the teacher model. In addition, the checkpoint-based saving structure used for both teacher and student models is considered part of the validation framework, since it supports best-model preservation, training continuity, and experimental reproducibility.

3.4 Interpretability Validation Criteria

In the evaluation of the interpretability module, the main focus is whether Grad-CAM activation maps reflect model decisions in a visually meaningful and spatially relevant manner. The main criteria in this context are whether the activation regions are concentrated on the nodule area, whether there is reasonable spatial overlap between the heatmap and the ground-truth mask, and whether over-expansion, incorrect spreading, or activation leakage into surrounding tissue occurs, especially in small nodules or nodules located near boundaries. In addition, the ability of the generated visualizations to make the decision process inspectable and to support the explanation of model behavior is accepted as one of the main validation criteria for this component. Since confidence values in Grad-CAM-based outputs may remain inflated, they are considered supportive indicators rather than fully reliable standalone evidence.

4. Dataset and Experimental Setup

The primary data source used in this study was the LIDC-IDRI dataset. Different components of the system used this dataset in different forms. In the segmentation pipeline, CT and SEG data were used together to generate anatomically aligned ROI-based samples. In the knowledge distillation-based classification structure, LIDC-IDRI data were used together with the LUNA16 subset and candidate nodule information in order to construct teacher and student model inputs. For the interpretability module, preprocessed image-mask pairs derived from the same dataset were used to generate Grad-CAM visualizations.

On the segmentation side, the preprocessing configuration was defined through dataset audit, CT-SEG matching, ROI extraction, and 2.5D sample generation. The initial audit covered the first 200 patients and identified 402 studies, 3,796 metadata records, 192 unique CT series, and 1,692 unique SEG objects. The final preprocessing setup used an ROI margin of 24 pixels, an output size of 192×192 , a 2.5D five-slice input representation, a positive-center-only sampling policy, and a patient-level split strategy. The main dataset and preprocessing settings for this component are summarized in Table 4.1..

Table 4.1. Dataset and preprocessing summary

Metric	Value
Patients analyzed	200
Studies	402
Metadata records	3796
Unique CT series	192
Unique SEG objects	1692
Average SEG per CT	8.81
Mean maximum nodule bbox	16 px
Median maximum nodule bbox	12 px
90th percentile bbox	33 px
95th percentile bbox	42 px
ROI margin	24 px
Output size	192×192
Input representation	2.5D (5 slices)
Sampling policy	Positive-center only
Split strategy	Patient-level

For the knowledge distillation-based classification component, data preparation included transforming candidate nodule locations from world coordinates to voxel coordinates, extracting $32 \times 32 \times 32$ 3D patches, clipping intensities to the $[-1000, 400]$ HU range, and scaling them to $[0, 1]$. For the student model, axial, coronal, and sagittal central slices were extracted from these 3D patches to form a three-channel 2.5D input.

The main input configuration, preprocessing settings, and distillation parameters are summarized in Table 4.2..

Table 4.2. Input configuration, preprocessing settings, and knowledge distillation parameters.

Category	Parameter	Value
Input Data	3D Patch Size (Teacher)	32x32x32
	2.5D Input (Student)	3 Channels (Axial, Coronal, Sagittal)
Preprocessing	Intensity Clipping (HU)	[-1000, 400]
Optimization	Learning Rate	0.001
Distillation	Alpha (α)	0.5

To support training continuity and reproducibility, checkpoint-based saving was used in both the segmentation and distillation stages. This structure preserved the main training state and enabled interrupted experiments to be resumed while keeping the best-performing model states available for evaluation.

5. Preprocessing Testing and Validation

The preprocessing stage was treated as a critical validation component because all subsequent segmentation, classification, knowledge distillation, and interpretability steps depend on the correctness of the generated inputs. For this reason, preprocessing was evaluated not only as a preparatory step, but also as a system component that required validation for each part of the pipeline. Since different system components required different input representations, preprocessing validation was organized into three parts: segmentation preprocessing, knowledge distillation preprocessing, and Grad-CAM/XAI preprocessing. The segmentation pipeline required CT–SEG matching, full-volume mask reconstruction, ROI extraction, and five-slice 2.5D image-mask generation. The knowledge distillation-based classification component required candidate-centered 3D patch extraction and tri-planar 2.5D student input preparation. The Grad-CAM-based interpretability module required inference-ready normalized inputs and spatially consistent heatmap overlays.

5.1 Segmentation Preprocessing Validation

For the segmentation component, preprocessing focused on converting heterogeneous LIDC-IDRI CT and SEG data into standardized ROI-based 2.5D image-

mask samples. Figure 5.1.1. illustrates the segmentation preprocessing workflow used to generate these samples and cache files.

The workflow starts with the LIDC-IDRI input, which includes patient-level CT, SEG, and related DICOM series. First, dataset audit and metadata inspection are performed to identify valid CT and SEG series and to check the required DICOM identifiers. Then, UID-based CT–SEG matching is applied to ensure that each SEG object is paired with the correct CT series. After matching, the sparse SEG frames are reconstructed in the CT slice space to obtain a full-volume mask aligned with the original CT volume.

Using the reconstructed mask, the nodule region is localized and an ROI is extracted by calculating a bounding box around the positive mask area with a 24-pixel margin. The extracted ROI is then resized to 192×192 pixels to standardize the input size for training. After ROI extraction, five-slice 2.5D samples are generated using the target center slice and its neighboring slices $[k-2, k-1, k, k+1, k+2]$. A positive-center-only sampling policy is applied, meaning that only samples whose center slice contains a positive nodule mask are included.

The generated samples are split at the patient level into training and validation sets to reduce data leakage between patients. Finally, the image-mask samples are saved as compact .npz cache files for each valid SEG object and used to train the segmentation model with DiceCELoss and checkpoint-based model saving.

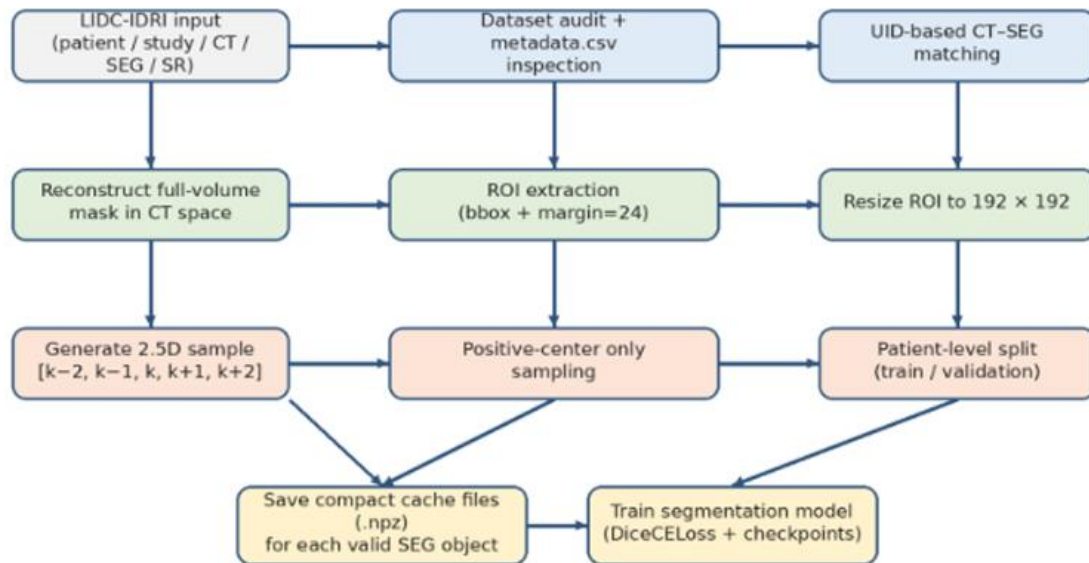


Figure 5.1.1. Segmentation preprocessing workflow used to generate ROI-based 2.5D samples and cache files.

The first validation step was to verify the reliability of the matching between CT and SEG data. In the proposed pipeline, CT and SEG data were matched using UID-based correspondence, after which SEG frames were mapped into CT slice space and converted into reconstructed full-volume masks. This step was essential because ROI

extraction and sample generation could only be considered valid if the anatomical relationship between the image and the annotation was preserved. The consistency of this transformation was checked visually through the generated preprocessing outputs.

The next validation focus was ROI extraction. After full-volume mask reconstruction, regions containing the nodule were cropped and resized. The purpose of this stage was to verify whether the extracted ROI preserved the lesion area while reducing irrelevant surrounding content. In addition, it was necessary to confirm that the resized outputs maintained the anatomical structure of the lesion and that the nodule region remained properly centered within the cropped sample. Figure 5.1.2. shows a representative segmentation preprocessing output, including the raw CT slice, CT slice with SEG annotation, preprocessed center slice, reconstructed mask, and image-mask overlay. This output was used to visually confirm that the reconstructed mask remained aligned with the lesion region after ROI extraction and resizing.

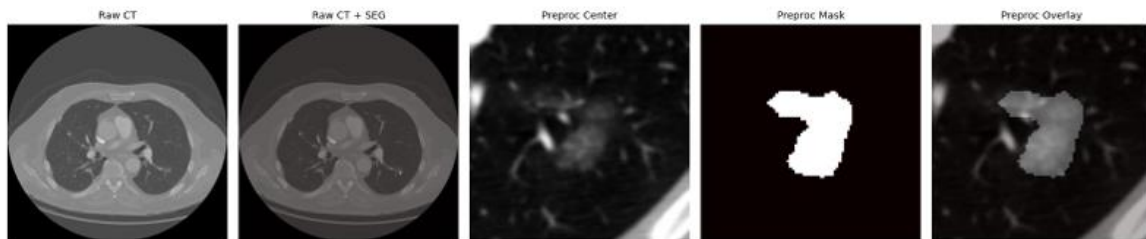


Figure 5.1.2. Example segmentation preprocessing output showing raw CT, raw CT with SEG annotation, preprocessed center slice, reconstructed mask, and image-mask overlay.

Another important aspect of segmentation preprocessing validation was the consistency of five-slice 2.5D sample generation. At this stage, the main question was whether the center slice and its corresponding mask remained anatomically aligned after sampling, and whether the neighboring slices preserved meaningful local context around the target region.

Figure 5.1.3. illustrates the five-channel 2.5D input representation used by the segmentation model. The input consists of five neighboring slices around the target slice, while the center-mask overlay is used to verify that the target mask corresponds to the correct center slice.

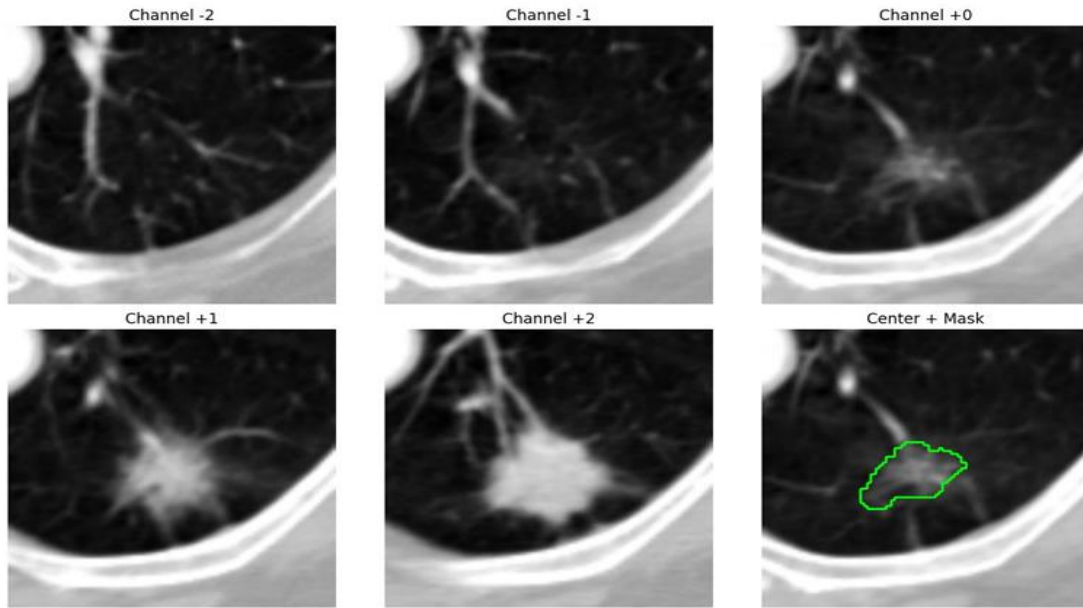


Figure 5.1.3. Five-channel 2.5D segmentation input representation with the center-mask overlay.

5.2 Knowledge Distillation Preprocessing Validation

For the knowledge distillation-based classification component, preprocessing focused on preparing consistent teacher and student model inputs. Candidate nodule locations were transformed from world coordinates to voxel coordinates, and $32 \times 32 \times 32$ 3D patches were extracted around candidate centers. The intensity values were clipped to the $[-1000, 400]$ HU range and scaled to $[0, 1]$.

The teacher model used the extracted 3D patches as volumetric inputs. For the student model, axial, coronal, and sagittal central slices were extracted from each 3D patch to form a three-channel 2.5D input. This preprocessing step was validated by checking whether the extracted 3D patches were centered on the candidate nodule location and whether the three student input slices represented the same candidate region as the teacher input. Therefore, the validation focus in this component was not image-mask alignment, but consistency between candidate coordinates, teacher inputs, and student inputs.

5.3 Grad-CAM / XAI Preprocessing Validation

For the Grad-CAM-based interpretability module, preprocessing focused on preparing inputs that were consistent with the trained classification model. The same normalized CT patches or ROI-based inputs used during classification inference were used to generate Grad-CAM heatmaps. This ensured that the explanations were produced from the same input representation used by the model during prediction.

After Grad-CAM heatmaps were generated, the activation maps were resized and overlaid on the corresponding CT slice. This step was validated by visually checking the spatial consistency between the CT slice, the expected nodule location, and the

generated activation map. The purpose of this validation was to confirm that the heatmap visualization could be meaningfully interpreted in relation to the anatomical region being analyzed.

Overall, the preprocessing testing and validation results show that the proposed pipeline produced input data suitable for different system components. For segmentation, preprocessing generated anatomically aligned ROI-based 2.5D image-mask samples. For knowledge distillation, preprocessing generated consistent 3D teacher inputs and tri-planar 2.5D student inputs. For Grad-CAM-based interpretability, preprocessing provided normalized inference inputs and spatially consistent overlay visualizations. In particular, the spatial coherence observed in Figure 5.3.1. and the efficiency findings of the pilot study provide a strong foundation for the later segmentation, classification, and interpretability stages.

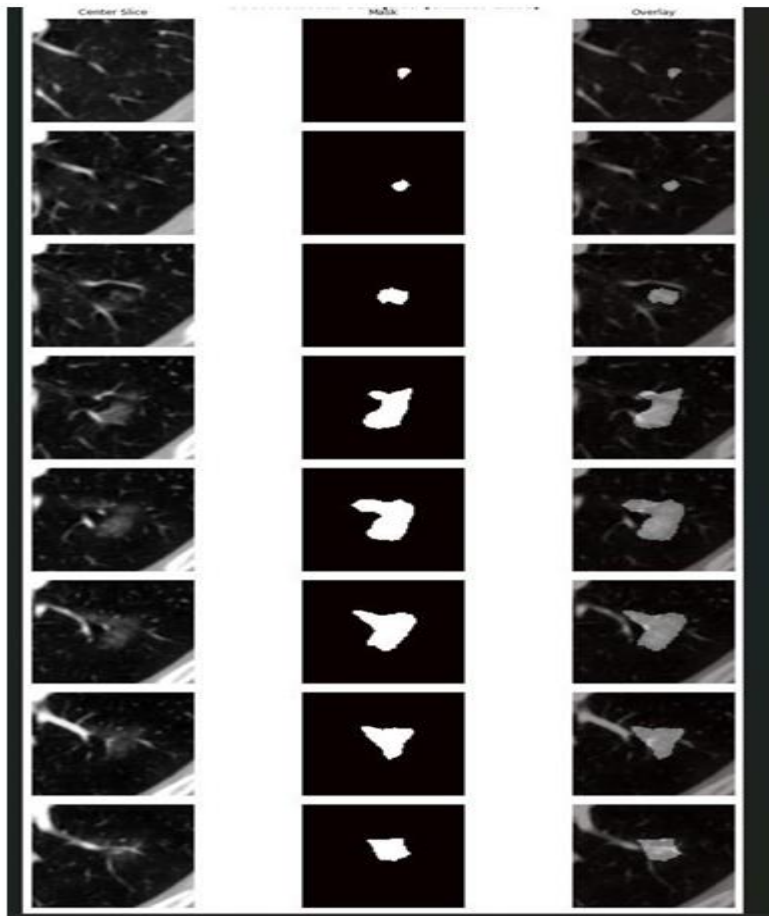


Figure 5.3.1. Overview of generated center-slice samples, masks, and overlays after segmentation preprocessing.

6. Segmentation Testing and Validation

The segmentation module was tested to evaluate whether it could learn meaningful spatial representations from the generated ROI-based 2.5D samples and whether it could predict the lesion region correctly. At this stage, the focus was no longer on the correctness of the data preparation pipeline, but rather on the model's training behavior, the quality of its mask predictions, and its generalization capacity on validation data. For this reason, segmentation validation was carried out through early-stage sanity checks, Dice and loss curves, final validation examples, and probability map analysis.

The first step of segmentation validation was an early-stage sanity check designed to verify whether the model had started learning in a meaningful way. The early validation predictions examined before the final results showed that the cached 2.5D inputs and the training pipeline were capable of producing non-random outputs. The early validation examples presented in Figure 6.1. provide an initial visual comparison between the input image, the ground-truth mask, and the predicted mask, thereby offering an intermediate confirmation of the learnability of the segmentation module. This stage was considered important because it showed that the model had at least started to distinguish the basic lesion structure.

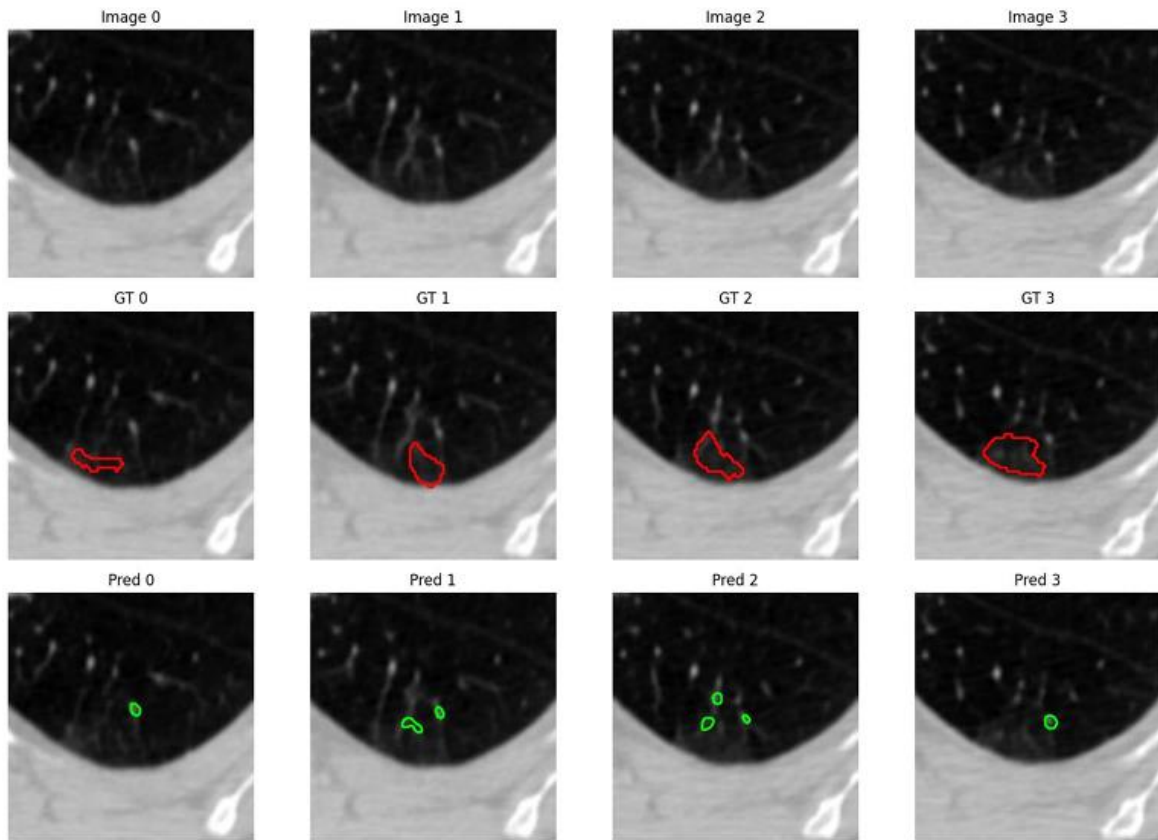


Figure 6.1. Early qualitative validation predictions showing input images, ground-truth masks, and predicted masks.

The quantitative evaluation of the segmentation module was performed using training and validation Dice scores together with loss curves. In the full-cache experiment, the convergence behavior of the model was monitored across epochs, and it was observed that training loss decreased steadily while validation loss followed a flatter trend after the early epochs. Similarly, the training Dice score increased, whereas the validation Dice score stabilized at a lower level. This behavior indicates that the model learned the segmentation task effectively, while also exhibiting a visible train–validation gap. These trends are clearly shown in Figure 6.2., while the main numerical results of the full-cache experiment are summarized in Table 6.1.

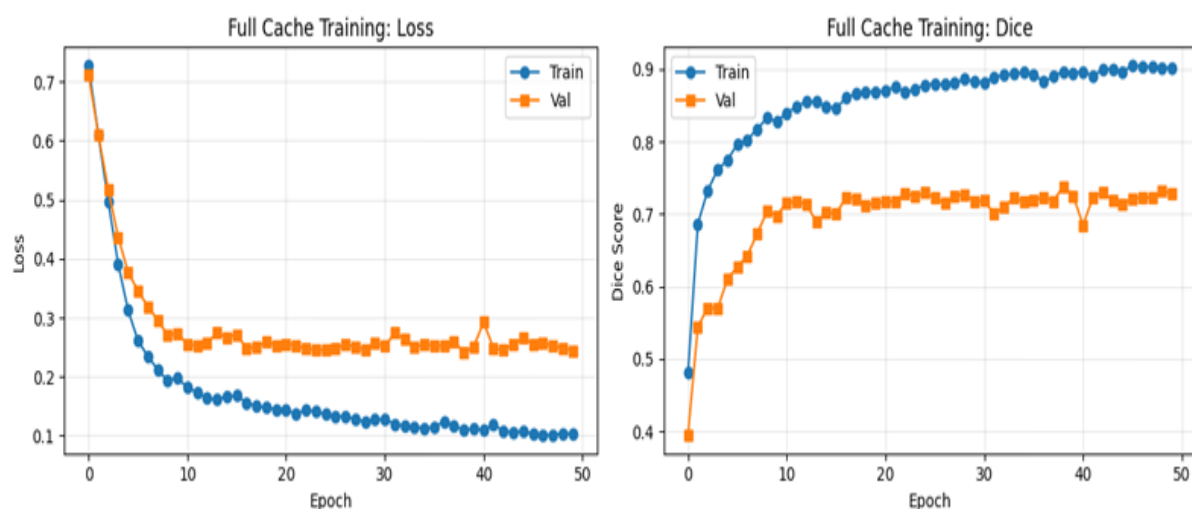


Figure 6.2. Full-cache training and validation loss and Dice curves.

Table 6.1. Final full-cache training results

Metric	Value
Usable cache files	187
Unique patients	187
Total samples	1148
Train patients	149
Validation patients	38
Train samples	933
Validation samples	215
Best checkpoint epoch	23
Best validation Dice	0.7366
Final training Dice	0.8757
Best training Dice	0.8759
Final training loss	0.1363

Final validation loss	0.2406
Loss function	DiceCELoss
Learning rate	0.001

The visual validation of segmentation outputs was performed through final validation examples. For this purpose, the input image, the ground-truth mask, the probability map, and the predicted mask were examined together in order to assess how well the model captured the main lesion region. The representative validation examples shown in Figure 6.3. indicate that the model was generally able to capture the main nodule area and to produce predictions that were visually consistent with the annotation. These visual outputs made it possible to assess the model’s pixel-level behavior beyond summary metrics such as Dice score.

Probability maps also played an important role in segmentation validation. These maps show where the model responded with higher confidence before thresholding and help explain to what extent the predicted boundaries follow the lesion structure. When the probability maps in Figure 6.3. are examined, the model is seen to concentrate mainly on the lesion region. However, some examples also show boundary leakage and small false-positive regions. These observations suggest that the model was successful in identifying the lesion region, but still has room for improvement in boundary precision and generalization.

Overall, the segmentation testing and validation results indicate that the proposed segmentation module was able to learn from ROI-based 2.5D samples, produce meaningful mask predictions, and localize the main lesion area with reasonable consistency. Figure 6.1. supports the early learnability check, Figure 6.2. shows the training and validation dynamics, and Figure 6.3. demonstrates the visual quality of the final mask predictions. Taken together, these components provide a strong validation basis for the segmentation module, while also showing that the train–validation gap, boundary errors, and small false-positive regions remain important areas for future improvement.

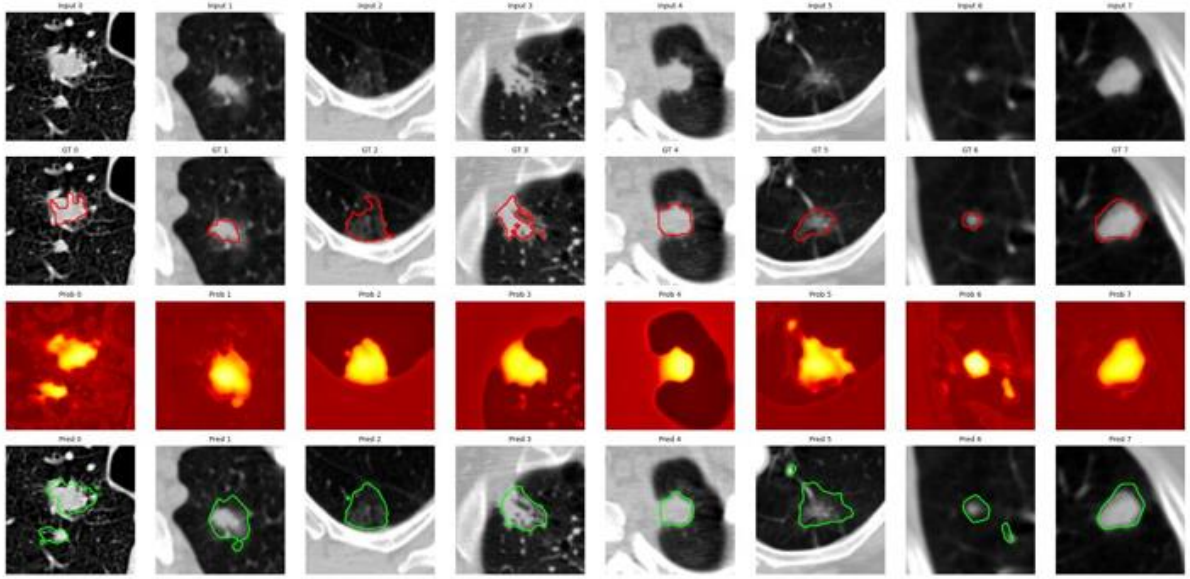


Figure 6.3. Validation probability maps and final predicted masks for representative full-cache samples.

7. Knowledge Distillation Testing and Validation

The knowledge distillation-based classification structure was tested to evaluate whether the volumetric information learned by the 3D teacher model could be transferred to a lighter 2.5D student model. At this stage, the validation focused on three main points: whether the teacher model provided a reliable reference, whether the student model showed measurable improvement during training, and whether the reported results were sufficient to evaluate classification performance in a medically meaningful way.

The first step of validation was the evaluation of the teacher model. In this framework, the 3D SegResNet was used as a high-capacity reference model that learned volumetric nodule-related features from 3D patches. Since the teacher model serves as the source of guidance for the student, its learned performance is an important part of the validation process. The teacher-side results used as the reference for knowledge distillation are consistent with the performance already reported earlier in Table 3, including the selected best checkpoint and the corresponding training and validation performance.

The second stage of validation focused on the student model. During student training, not only the ground-truth labels but also the soft-target outputs generated by the teacher model were used. Through this structure, the student model was expected to approximate the teacher's volumetric knowledge using a lighter 2.5D input composed of axial, coronal, and sagittal views. The reported results show that the student model improved during training, with accuracy increasing from 74.09% at the beginning of the distillation process to 87.28% at the final evaluation stage. In addition, the final training accuracy reached 90.41%, while the distillation loss decreased from 0.5801 to 0.2821.

However, these reported values are not sufficient on their own to fully validate the classification model. Since the dataset is class-imbalanced and the task is medically sensitive, accuracy alone may be misleading. In particular, recall/sensitivity is critical because false negative predictions may correspond to missed nodule cases. Therefore, the final test evaluation should also include recall, precision, F1-score, ROC-AUC, specificity, and a confusion matrix. These metrics were not included in the current experimental output and should be calculated from the final test predictions and predicted probabilities before making a stronger conclusion about clinical reliability.

Another important aspect of validation was to examine whether the 2.5D student model benefited from the teacher-guided distillation process. The available results show that the student accuracy increased from 74.09% at the beginning of training to 87.28% at the final evaluation stage, while the distillation loss decreased from 0.5801 to 0.2821. These results suggest a positive training trend under the guidance of the teacher model. However, this conclusion is based only on the reported accuracy and loss values. Since additional classification metrics such as recall, F1-score, ROC-AUC, and confusion matrix analysis were not reported, the clinical reliability of the student model cannot be fully evaluated.

Similarly, although the 2.5D student model is expected to be computationally lighter than the 3D teacher model because it uses tri-planar 2D inputs instead of full 3D patches, no direct computational cost analysis was performed. Inference time, memory usage, FLOPs, parameter count, or millisecond-level runtime measurements were not reported. Therefore, the efficiency advantage should be interpreted as an architectural expectation rather than an experimentally verified result.

8. Interpretability Testing and Validation

The interpretability module was tested to evaluate to what extent the spatial regions influencing the model’s decisions could be reflected in a visually meaningful way. At this stage, the main focus was whether the Grad-CAM-based activation maps were spatially aligned with the nodule region, whether model attention was concentrated on clinically meaningful areas, and whether the generated visualizations made the decision process inspectable. For this reason, validation was carried out through the suitability of the Grad-CAM setup, target layer selection, visualization outputs, activation–mask overlap, confidence score behavior, and interpretability limitations.

First, the technical setup of the Grad-CAM module was evaluated from a validation perspective. In this structure, the input data were taken from preprocessed .npz image–mask pairs, and the midpoint sample together with the midpoint contextual channel was selected to make the data suitable for 2D visualization. The image and mask were then resized, the image was normalized, and the input was adapted to match the channel structure expected by the model. The main configuration parameters used for Grad-CAM are summarized in Table 8.1. This table presents the input tensor

dimensions, target resolution, normalization range, model structure, the GradCAMPlusPlus method, target layer selection as `model.conv_final`, and the confidence method together. In this way, the interpretability module is shown to operate with a setup that is not only visual but also technically defined and inspectable.

The second aspect of validation was the examination of the spatial alignment between the activation maps and the nodule region. For this purpose, the raw CT slice, the ground-truth mask, and the Grad-CAM heatmap were evaluated together. In the example shown in Figure 8.1, the activation is concentrated in the core of the nodule, decreases from the center toward the periphery, and remains at a lower level in the surrounding lung tissue. This behavior was considered positive from an interpretability perspective, because it indicates that the model directs its attention to lesion-related regions rather than distributing it randomly across the entire image. In addition, this example demonstrates a strong spatial overlap between the activation region and the ground-truth mask.

However, interpretability evaluation was not based only on successful examples. It was also observed that, in small nodules or lesions located near boundaries, the activation region could extend beyond the ground-truth mask and spread into surrounding tissue. This behavior is clearly visible in Figure 8.3. In that example, the Grad-CAM heatmap generated for a small nodule still covers the nodule region, but also extends into adjacent tissue. Such incorrect spreading or over-activation patterns suggest that, especially for small targets, contextual features captured by deeper layers may go beyond the lesion boundary. Therefore, while Grad-CAM heatmaps provide valuable visual explanations of model attention, they were not assumed to ensure exact boundary accuracy in every case. This was considered an important limitation of the interpretability module.

Another element considered in interpretability validation was the behavior of the confidence score. When the confidence method and the observed confidence range reported in Table 8.1 are considered together, it can be seen that the model confidence values often remain at very high levels. The report notes that this may be related to the segmentation model producing dense probability maps and the confidence score being derived from the full output area. Therefore, these scores were not treated as fully reliable standalone indicators at the nodule level. This observation shows that visual explanations should be interpreted not only by looking at the activation area, but also by considering how the confidence definition is constructed.

Overall, the interpretability testing and validation results show that the Grad-CAM module provides a useful qualitative tool for inspecting model attention. Table 8.1 presents the technical setup and visualization parameters of the module, while Figure 8.1 and Figure 8.2. show representative Grad-CAM outputs with different localization behaviors. Figure 8.1. demonstrates a visually stronger overlap between the activation heatmap and the annotated nodule region, whereas Figure 8.2. illustrates broader activation around a small nodule.

However, the Grad-CAM evaluation in this report is based on visual inspection only. Quantitative localization metrics such as IoU (Intersection over Union) and Dice Score were not calculated for the Grad-CAM heatmaps. Therefore, these outputs should be interpreted as qualitative evidence of model attention rather than definitive localization performance. To make a stronger validation claim, future work should threshold the Grad-CAM heatmaps and compare them with the ground-truth masks using IoU and Dice Score.

In this sense, the interpretability module increases the inspectability of the system's decision-making process, but it should be used as a supportive validation tool rather than a definitive measurement of model correctness.

Table 8.1. GradCAM Pipeline Parameters and Configuration Values

Category	Parameter	Value
Input Data	Raw image tensor shape	(8, 5, 192, 192)
	Raw mask tensor shape	(8, 192, 192)
	Selected slice index (image)	batch=4, channel=2 (midpoints)
	Selected slice index (mask)	batch=4 (midpoint)
Preprocessing	Target spatial resolution	224 x 224 px
	Normalization range	[0, 1]
	Normalization epsilon	1e-8
	Binarization threshold (mask)	0.5
	Replicated input channels	5
Model (SegResNet)	Spatial dimensions	2D
	Input channels	5
	Output channels	1
	Initial filters	8
	Encoder blocks	(1, 2, 2, 4)
	Decoder blocks	(1, 1, 1)
Grad-CAM	CAM method	GradCAMPlusPlus
	Target layer	model.conv_final
	Target class	MultiSliceSegmentationTarget (category=0)
	Gaussian blur kernel	(7, 7), sigma=0
	Colormap	Custom ListedColormap (jet-based, n=256)
	Colormap threshold	0.3
	Alpha range	0.1 to 0.8
	CAM map inversion	Yes (1.0 - cam_map_norm)
	CAM overlay opacity (alpha)	0.6
	Colorbar fraction	0.046
Results	Colorbar padding	0.04
	Confidence method	Mask-restricted (predicted > 0.5), fallback to max

	Observed confidence range	~96%
	Output figure dimensions	18 x 6 inches

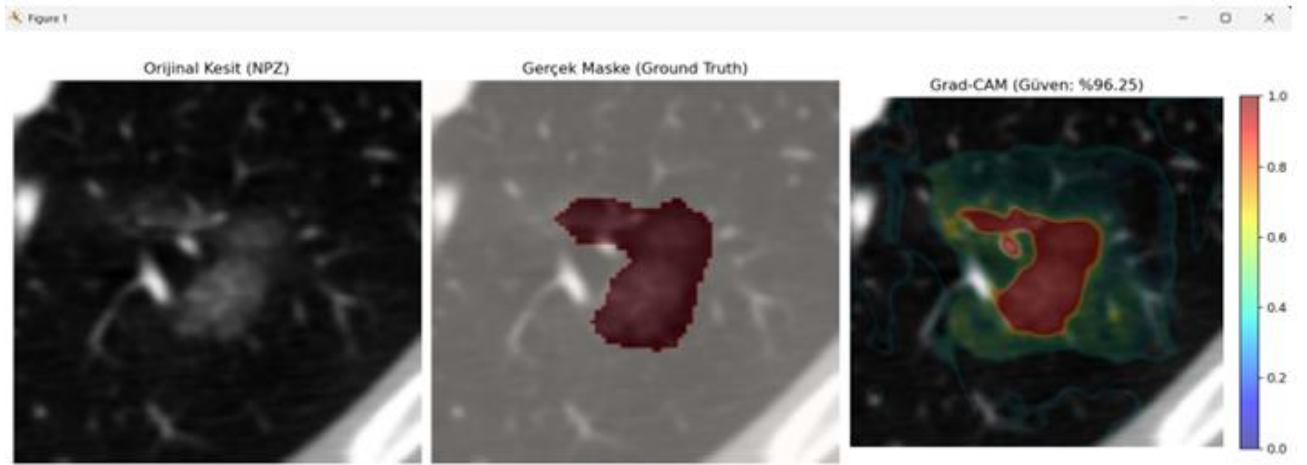


Figure 8.1. Grad-CAM on LIDC-IDRI NPZ Sample (Confidence: 96.25%)



Figure 8.2. Grad-CAM on LIDC-IDRI NPZ Sample (Confidence: 96.56%)

9. Limitations and Threats to Validity

Although the results obtained in this study indicate that the main components of the system are functional, there are several limitations and threats to validity that should be considered when interpreting the findings. These threats appear at different levels of the system, ranging from preprocessing to segmentation and classification performance, as well as the interpretability module. For this reason, it is important to state not only the strengths of the results, but also the conditions under which they remain limited.

The first major limitation is the train–validation gap observed in the segmentation module. Although training loss decreased steadily and the training Dice score increased, the fact that validation loss flattened earlier and validation Dice stabilized at a lower level suggests that the generalization capacity of the model remained more limited than its training performance. This indicates that, while the model was learnable, it still remained susceptible to overfitting and may require stronger regularization or more diverse data support.

A second limitation is the presence of boundary leakage and small false-positive regions in some segmentation outputs. The inspection of probability maps and predicted masks showed that the model was generally able to capture the main lesion area, but also produced predictions that extended beyond lesion boundaries in certain examples. This indicates that pixel-level localization performance was not equally stable in all cases and that boundary precision still requires improvement. Therefore, the achieved Dice performance should not be interpreted as implying perfect spatial accuracy in every sample.

There are also important threats to validity on the knowledge distillation side. Although there is evidence that the student model benefited from the teacher model, it is not certain to what extent the lighter 2.5D structure can fully preserve the volumetric information learned by the 3D teacher model. The tri-planar 2.5D input structure provides axial, coronal, and sagittal views of the candidate region, which can offer more contextual information than a single 2D slice. However, because the student model does not process the full $32 \times 32 \times 32$ volume directly, it should be interpreted as an approximation of 3D context rather than a complete replacement for the volumetric representation used by the teacher model. For this reason, the performance achieved by the student model should be interpreted as an efficient approximation rather than a complete equivalent of the teacher model’s representational capacity.

Another limitation concerns the behavior of the confidence score. In the Grad-CAM evaluations, the reported confidence values remained very high, with the presented samples showing confidence scores around 96% (for example, 96.25%, 96.05%, 96.56%, and 96.37%). However, these confidence values should be interpreted carefully because they were not supported by additional localization metrics such as IoU or Dice Score. Therefore, high confidence does not necessarily indicate that the activation map perfectly overlaps with the ground-truth nodule mask. Future evaluation should include mask-restricted confidence estimation and quantitative overlap metrics to make the confidence score more clinically meaningful. This was suggested to be related to dense probability maps and to the fact that confidence was derived from the full output region. Therefore, these scores should not be treated as fully reliable standalone decision indicators at the nodule level. This becomes especially important in the interpretation of visual explanations, because an apparently meaningful activation pattern does not necessarily correspond to a fully trustworthy numerical confidence estimate.

The interpretability module itself is also not a flawless explanation mechanism. Although Grad-CAM activation maps were generally focused on the nodule region in many cases, activation was also observed to extend beyond the ground-truth mask in small nodules or lesions located near boundaries. This suggests that, while visual explanations are useful, they do not always provide exact boundary precision and may be influenced by contextual activations around small targets. For this reason, Grad-CAM outputs should be interpreted as supportive visual evidence rather than definitive proof of model reasoning.

Another important validity concern is that a significant portion of the evaluation depends on visual inspection. Preprocessing validation and ROI consistency checks were supported by specific visual outputs, including raw CT, CT+SEG, preprocessed center slice, reconstructed mask, and image-mask overlay examples. These visualizations were used to check whether the reconstructed mask remained spatially aligned with the nodule region after ROI extraction and resizing. Early segmentation sanity checks were supported by qualitative prediction examples showing input images, ground-truth masks, and predicted masks. Grad-CAM interpretation was supported by three-panel visualizations comparing the original CT slice, the ground-truth mask, and the Grad-CAM heatmap. Although this approach is valuable for understanding model behavior, such qualitative observations may remain sensitive to the selected examples and may not fully represent the entire data distribution. Therefore, visual validation should not be interpreted as sufficient evidence on its own. In this report, visual examples were used mainly for qualitative inspection of preprocessing consistency, segmentation behavior, and Grad-CAM attention. Since quantitative indicators such as IoU, recall, F1-score, and ROC-AUC were not fully reported for all components, these visual results should be considered supportive evidence rather than definitive proof of model reliability.

Finally, it should be noted that the experimental results are limited to the current data subsets and the present experimental setup. On the preprocessing and segmentation side, the dataset audit was conducted on the first 200 patients, and the final full-cache experiment was based on 187 usable cache files. While these experiments demonstrate the feasibility of the proposed system, it remains necessary to verify whether similar findings would be maintained across broader, more diverse, and independent data groups. This does not invalidate the reported results, but it does represent a clear limitation in terms of generalizability.

Overall, these limitations do not invalidate the contribution of the system; rather, they define the context in which the results should be interpreted. The correctness of the preprocessing pipeline, the learnability of the segmentation module, the functional value of the distillation structure, and the explanatory contribution of Grad-CAM all represent positive outcomes. At the same time, the generalization gap, the reliability of confidence scores, activation overflow in small nodules, and the dependence on visual evaluation remain open areas for future improvement and more comprehensive validation.

10. Report Generation and Turkish NLP Testing and Validation

10.1. Testing Results Summary for the Report Generation Module

The testing process confirmed that the nodule report generation module operates in accordance with the Fleischner Society 2017 follow-up recommendations. It was observed that the system generates appropriate follow-up outputs for both solid and subsolid nodules by considering nodule size, single versus multiple nodule status, and available clinical risk factor information. When clinical risk information is available, the module differentiates the recommendation according to the patient's risk profile. When such information is absent, the system continues report generation using a safe default flow without interrupting the clinical reporting process.

The validation results showed that the rule-based reporting logic correctly reflects the expected clinical recommendation pathways. In particular, solid nodules smaller than 6 mm were handled with no routine follow-up in low-risk cases and optional 12-month CT recommendations in high-risk cases. For solid nodules in the 6–8 mm range, follow-up intervals were correctly generated within the 6–12 month and 18–24 month windows. For solid nodules larger than 8 mm, the system produced recommendations involving short-term CT follow-up and further evaluation such as PET/CT or biopsy. For subsolid nodules, both ground-glass and part-solid pathways were correctly distinguished, and the corresponding short-term confirmation CT schedules were applied properly. In multiple subsolid cases, the report generation logic reflected a management strategy based on the most suspicious nodule.

Because the module operates with rule-based logic, identical inputs consistently produce identical clinical recommendations. This characteristic increases testability, traceability, and reproducibility of the system. Furthermore, when clinical risk data are incomplete, the module still generates safe and explainable report text without interrupting the workflow.

Overall, the Fleischner 2017-based reporting logic was successfully validated in the test phase, and the critical follow-up recommendations were generated in the correct classes and within the correct time windows. Therefore, the report generation module satisfied the defined test acceptance criteria. The main test scenarios and observed outputs of the rule-based report generation module are summarized in Table 10.1.1.

Table 10.1.1. Validation Results of the Rule-Based Nodule Report Generation Module

Test Scenario	Expected Outcome	Observed Outcome	Status
Single solid nodule, 5 mm, low-risk patient	No routine follow-up should be recommended	The generated report included the equivalent of “no routine follow-up”	Passed
Single solid nodule, 5 mm, high-risk patient	Optional CT at 12 months should be recommended	The generated report correctly included an optional 12-month CT recommendation	Passed
Part-solid nodule, 7 mm	Short-term confirmation CT at 3–6 months and subsequent annual follow-up should be recommended	The generated report correctly followed the part-solid management pathway	Passed
Ground-glass nodule, 7 mm	Confirmation CT at 6–12 months and long-term follow-up should be recommended	The generated report correctly followed the ground-glass management pathway	Passed

10.2 Clinical Rules Verified During Testing

The following clinical rules were verified during the test process:

- **Solid nodule, < 6 mm:** No routine follow-up is generated in low-risk patients, while an optional 12-month CT recommendation is produced in high-risk cases.
- **Solid nodule, 6–8 mm:** Follow-up CT recommendations are generated within the 6–12 month and 18–24 month intervals.
- **Solid nodule, > 8 mm:** Short-term CT and further evaluation options such as PET/CT or biopsy are recommended.
- **Subsolid nodule:** Ground-glass and part-solid lesions are handled separately, and the appropriate 3–6 month or 6–12 month confirmation CT workflows are applied.
- **Multiple subsolid nodules:** Management is reflected according to the most suspicious nodule.
- **Clinical risk factors:** Smoking history, family/genetic history, and prior malignancy history shift the recommendation toward the high-risk pathway when present.

10.3 BERTurk Integration Test Results

The BERTurk-based NLP integration was evaluated to verify Turkish clinical text analysis for extracting risk factor signals and nodule morphology cues. The integration was designed as an auxiliary signal provider within the reporting pipeline, while the final follow-up decision remained deterministically controlled by the Fleischner-based rule engine.

The purpose of this test phase was to verify whether Turkish-language clinical text could be processed reliably and whether extracted risk-related information could be safely incorporated into the reporting workflow without affecting the integrity of the structured clinical data. The tests also aimed to confirm that the system could continue operating through a fallback mechanism if the NLP service became unavailable.

The results demonstrated that when the NLP service was reachable, the backend successfully retrieved patient risk profile signals and nodule-type indicators from Turkish clinical input. These outputs were stored as separate supportive fields, such as extracted risk factors, inferred morphology type, and confidence score, while preserving the original structured report fields. The report generation flow continued successfully regardless of whether NLP output was available. The evaluation focused on whether Turkish NLP outputs could support the reporting workflow without disrupting the deterministic rule-based recommendation mechanism.

From a quality perspective, the NLP integration showed high reliability because service continuity was preserved through fallback behavior. Data integrity was also maintained since NLP outputs were stored in supportive fields without overwriting the core clinical findings. In terms of clinical explainability, the system remained transparent because the final follow-up recommendation continued to be generated by the deterministic rule-based engine. Performance was considered satisfactory in hybrid mode, although additional caching or deployment optimization would be beneficial if the full classifier inference mode is used in production.

As a result, the BERTurk integration met the test acceptance criteria. The system was able to add Turkish clinical text understanding to the reporting pipeline in a safe manner while preserving the consistency and explainability of the rule-based follow-up decision mechanism. Table 10.3.1. summarizes the validation areas, test conditions, expected results, observed results, and final status of the BERTurk-based Turkish NLP integration.

Table 10.3.1. Validation Summary of the BERTurk-Based Turkish NLP Integration

Validation Area	Test Condition	Expected Result	Observed Result	Status
Risk factor extraction	Turkish clinical note includes smoking history	The NLP module should detect smoking as a risk-related signal	Smoking history was extracted and saved as a supportive risk factor signal	Passed
Nodule morphology extraction	Turkish clinical note includes a part-solid nodule expression	The NLP module should identify the morphology as part-solid	The part-solid morphology signal was extracted and transferred to the reporting workflow	Passed
Fallback mechanism	NLP service is unavailable during report generation	The system should continue report generation using the rule-based workflow	Report generation continued without interruption through the fallback rule-based flow	Passed
Data integrity	NLP output is available together with structured clinical data	NLP outputs should not overwrite the original structured clinical fields	Extracted NLP signals were stored in separate supportive fields, while core clinical fields were preserved	Passed

10.4 Quality Evaluation of the BERTurk Integration

Evaluation Criterion	Assessment	Explanation
Reliability	High	The system preserves service continuity through fallback behavior when the NLP service is unavailable

Evaluation Criterion	Assessment	Explanation
Data Integrity	High	NLP outputs are stored in auxiliary fields and do not overwrite core clinical data
Clinical Explainability	High	Final follow-up recommendations remain fully traceable because they are produced by the rule-based engine
Performance	Good in hybrid mode	Stable in default hybrid mode; initial latency exists in full classifier mode due to model download and loading time

10.5 Final Evaluation

The test results confirm that the report generation module and the BERTurk-assisted Turkish NLP integration operate consistently within the designed system architecture. The rule-based report generation component successfully produces Fleischner 2017-compliant follow-up recommendations, while the BERTurk integration adds supportive clinical text understanding without altering deterministic clinical decision logic. Taken together, these results indicate that the system satisfies the defined acceptance criteria in terms of correctness, continuity, and explainability.

11. Conclusion

This testing and validation report evaluated a multi-stage lung nodule analysis pipeline developed on the LIDC-IDRI dataset, including preprocessing and ROI-based 2.5D segmentation, knowledge distillation-based classification, and Grad-CAM-based interpretability. The overall findings indicate that the proposed system established a technically coherent workflow from data preparation to prediction and visual explanation.

The preprocessing stage showed that heterogeneous CT and SEG data could be transformed into anatomically aligned and visually consistent ROI-based 2.5D samples. The segmentation module demonstrated meaningful learning behavior through representative validation outputs and quantitative performance trends. The knowledge distillation framework showed that a lighter 2.5D student model could benefit from the volumetric information learned by the 3D teacher model, supporting a practical balance between representational strength and computational efficiency. In addition, the Grad-CAM module contributed an interpretable layer by making the model's spatial focus visually inspectable in relation to the nodule region.

Overall, the study provides a solid validation basis for the proposed pipeline and shows that the system is functional, interpretable, and suitable as a foundation for further development. By combining anatomically aware preprocessing, ROI-based 2.5D learning, teacher–student knowledge transfer, and visual interpretability, the pipeline offers a structured framework for AI-assisted lung nodule analysis. Further improvements can focus on strengthening generalization, refining confidence estimation, and extending validation on broader datasets.